Exploratory Evaluation of a Transcriptomics-Based Drug Repurposing Pipeline to Identify Compounds for Paclitaxel-Induced Peripheral Neuropathy

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Background and Purpose

Peripheral neuropathy is a major dose-limiting toxicity of paclitaxel, a commonly used drug to treat breast cancer. Paclitaxel-induced peripheral neuropathy (PIPN) has a significant negative impact on survivors' employment status, functional status, and quality of life (QOL). However, treatments for PIPN have limited efficacy. The molecular complexity and multifactorial nature of PIPN pose unique challenges for the development of effective therapies and suggest the need for precision health that considers individual variability in the transcriptome. While computational drug discovery approaches have been used to identify new treatments for a variety of medical conditions, and are promising for identifying effective drugs for PIPN, all of the work done to date has used non-computational approaches with limited success. Integrating the gene expression signature of this adverse sequelae with drug expression data provides an unprecedented opportunity to leverage precision health in the context of computational drug repositioning.

The goals of this preliminary study are to identify targets for treatment with transcriptomics, leverage these data and apply a computational drug-repurposing pipeline to identify single-agent therapies from existing drugs based on expression reversal, and to evaluate the findings of other studies.

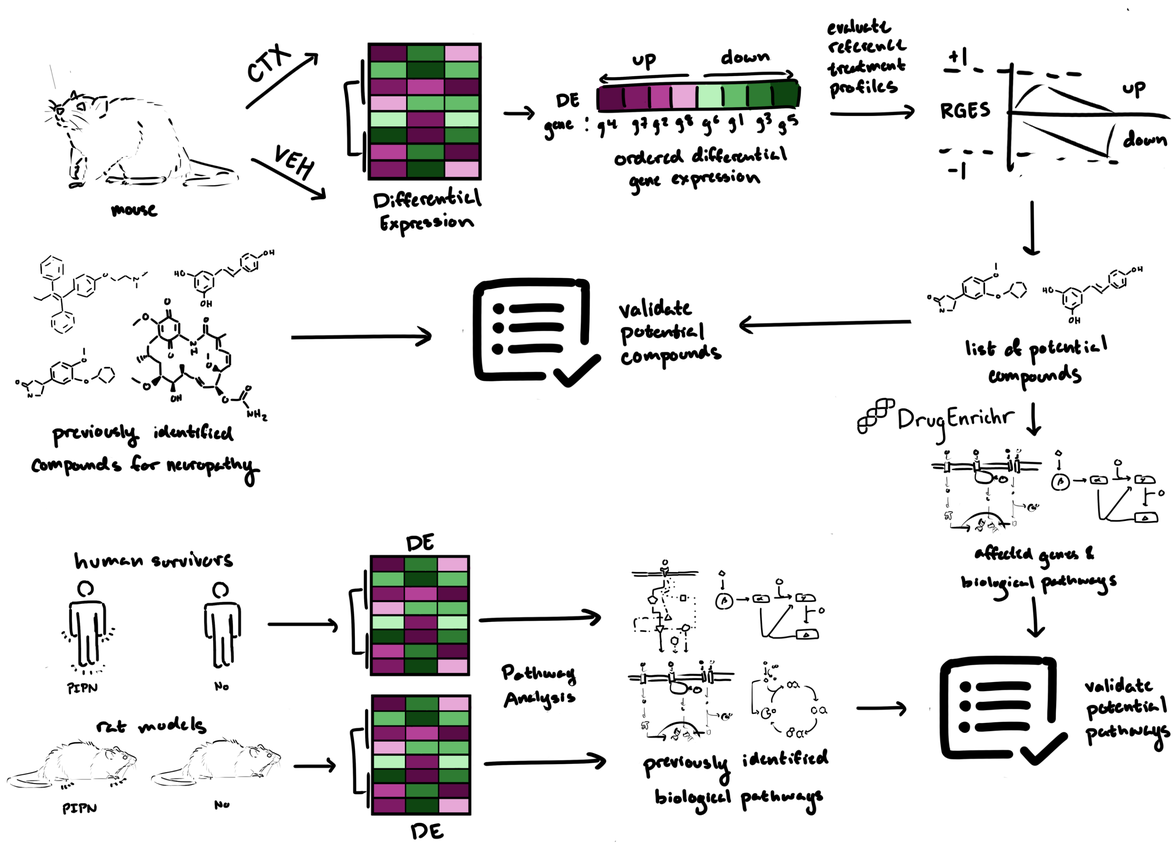

Figure 1. Overview of the study to identify potential compounds to treat paclitaxel-induced peripheral neuropathy. CTX: chemotherapy, DE: differential expression, PIPN: paclitaxel-induced peripheral neuropathy, RGES: reverse gene expression score, VEH: vehicle

Methods

Bulk gene expression RNA-seq data were obtained from a publicly available study (GSE113941) of nine pools of mice dorsal root ganglia (DRG) after treatment with paclitaxel (n=5) in a dose that was sufficient to induce PIPN compared to non-treated mice (n=4). Differential gene expression was evaluated using gene expression counts with edgeR. A PIPN signature was then created from differentially expressed genes ($p<0.01$, $logFC > 1.0$).

A transcriptomics-based drug repurposing pipeline was then applied to the signature to identify PIPN-drug pairs with opposite transcriptional effects (Chen 2017). Reversal scores were calculated for each drug in the CMap dataset (1000 small-molecule drugs) and 66,510 LINCS signatures. Significance was assessed using permutation analysis (n=10,000). Drug hits with a nominal p-value < 0.005 and reversal scores < 0 (indicating signature reversal) were examined. Candidate drugs were compared to 11 previously identified compounds to treat PIPN.

KEGG pathways targeted by these drugs were evaluated by enrichment analysis (FDR < 0.01) using DrugEnrichr. These targeted pathways were compared to those perturbed in pathway impact analysis of bulk gene expression RNA-seq data isolated from DRG in a rat model of acute PIPN (GSE160543) and from peripheral blood in breast cancer survivors with and without persistent PIPN (Kober 2017).

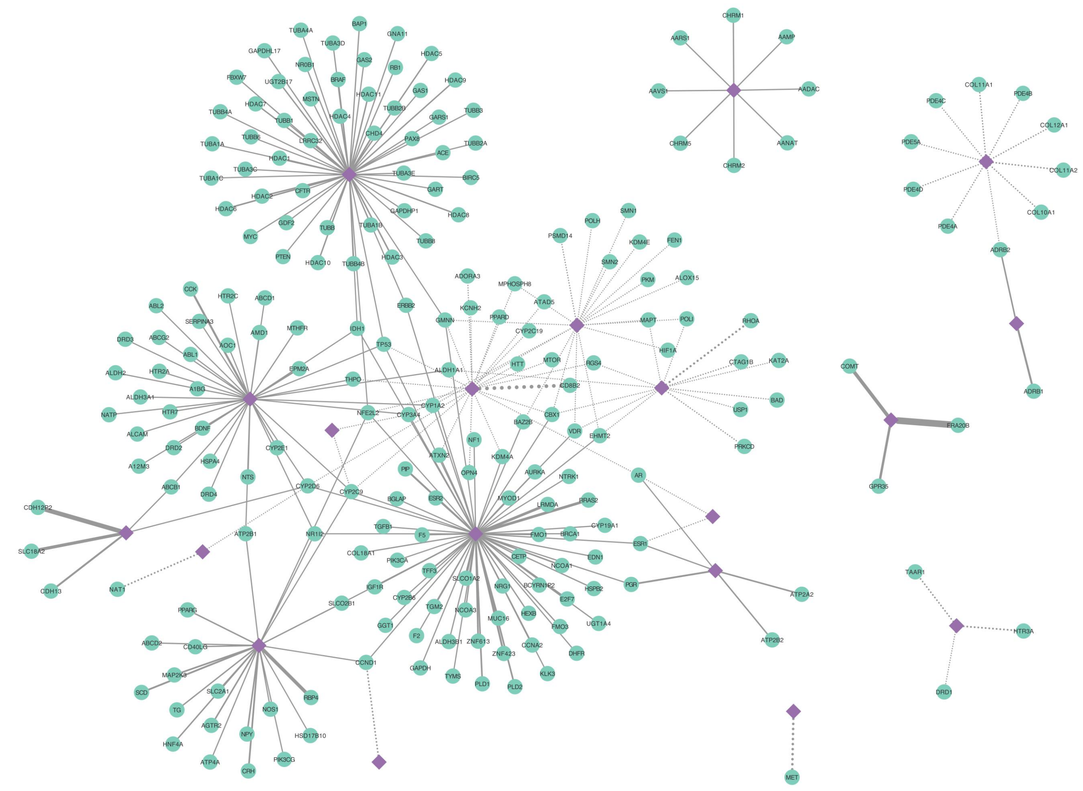

Figure 2. Drug-target interaction network of the top 20 candidate compounds identified with DGIdb. Edges are scaled by the interaction score between drugs (purple diamond nodes) and genes (green circle nodes). Edges lines are depicted for regulatory approval (solid) or not (dotted).

Results and Discussion

A transcriptomic profile of 26 landmark genes was identified from the differential expression analysis (n=10 up, n=16 down). From this signature, 442 drug signatures were identified. Four drugs (resveratrol, geldanamycin, rolipram, and tamoxifen) have been previously identified as demonstrating treatment efficacy for peripheral neuropathy, suggesting this approach is effective in identifying compounds to treat PIPN. Notably, the most commonly used drug for PIPN, duloxetine, was not identified as a candidate drug. Given duloxetine has limited efficacy, the drugs identified using this method of gene expression reversal may offer promising alternatives.

Of the identified 135 KEGG biological pathways enriched for these candidate drugs, 120 (e.g., Focal adhesion, HIF-1 signaling, Cellular senescence) were also identified in a rat preclinical model of acute PIPN and 36 (e.g., mitophagy, HIF-1 signaling, ferroptosis, axon guidance) were previously identified by our research team as potential targets in a series of studies of 50 breast cancer survivors (Kober 2018, Kober 2019, Miaskowski 2019, Kober 2020). These overlap suggest a consistent molecular pattern in the targets of these candidate drugs.

Conclusions

In this exploratory study, using a transcriptomics-based drug repurposing pipeline, we identified potential compounds to treat PIPN. These findings suggest this approach may provide a productive approach to drug discovery to treat this severe adverse side effect of neurotoxic chemotherapy. Future studies are needed to evaluate the effects of these compounds in *ex vivo* and *in vivo* models. Future studies are needed to confirm these findings in preclinical model and in patients receiving treatment.

Acknowledgements

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